

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)   |  | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | salt petre / KNO <sub>3</sub> / potassium                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |     | 1   | 1     | 1     |      |
|          | (b) | (i)   |  | so the <u>ions</u> can <u>move</u><br><br>accept reference to the lead/Pb <sup>2+</sup> and chloride/Cl <sup>-</sup> ions                                                                                                                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       | 1    |
|          |     | (ii)  |  | (because the chloride ions are) negatively charged (1)<br><br>opposite charges attract (1)                                                                                                                                                                                                                                                                                                                                                                                             | 1               | 1   |     | 2     |       | 2    |
|          |     | (iii) |  | (the lead ions) <u>gain electrons</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 | 1   |     | 1     |       |      |
|          |     | (iv)  |  | $2\text{Cl}^- + 2\text{e}^- \rightarrow 2\text{Cl}$ <input type="checkbox"/><br><br>$\text{Cl}^- + \text{e}^- \rightarrow \text{Cl}^{2-}$ <input type="checkbox"/><br><br>$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ <input checked="" type="checkbox"/><br><br>$\text{Cl}^- + \text{e}^- \rightarrow \text{Cl}$ <input type="checkbox"/><br><br>$\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ <input type="checkbox"/><br><br>accept any correct indication of answer |                 |     | 1   | 1     |       |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |          |          |           |          |          |
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|          |  |  |  | <p><b>Indicative content</b></p> <ul style="list-style-type: none"><li>• near a dock / coast – to import the aluminium ore from overseas – because there is no aluminium ore in the UK</li><li>• near a power station – to negotiate a cheap electricity supply – because of the large amounts used and the high costs</li><li>• railway links – to be able to easily transport the aluminium</li><li>• road links – to be able to transport the aluminium and for access for employees</li><li>• housing area – available workforce nearby</li><li>• distance from homes – reduce impact of noise pollution</li></ul> <p><b>AO1 – recalling factors important to location (3)</b><br/><b>AO2 – application to given example (3)</b></p> <p><b>5-6 marks</b><br/>Five or six of the factors are identified and correctly justified, including the need to be near the dock to import the aluminium ore from overseas and the power station to get sufficient electrical power; explanations are detailed and clearly justify why each factor is important for the location<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i></p> <p><b>3-4 marks</b><br/>Two or three of the factors are identified and some attempt at justifying their importance; explanations are reasonably detailed but might not justify why the factor is important to the process<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i></p> <p><b>1-2 marks</b><br/>One or two factors listed but little/no attempt at justifying their importance; there is little or no attempt at any explanation<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i></p> <p><b>0 marks</b><br/><i>No attempt made or answer worthy of any credit.</i></p> | 3               | 3        |          | 6         |          |          |
|          |  |  |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>6</b>        | <b>5</b> | <b>2</b> | <b>13</b> | <b>1</b> | <b>3</b> |